Naive Bayes Foundations: Mutually Exclusive Events

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Independent and mutually exclusive events.

1. Overview

Key point: Two events are **mutually exclusive** (G-1286) when they cannot happen together: P(A ∩ B) = 0. Mutual exclusivity is a different idea from independence, and the two are easily confused.

Independent events (G-934) can happen together, but one does not affect the other. **Mutually exclusive** events are different: they can never happen together. The two ideas are easy to mix up, so this Note puts them side by side.

2. The definition

Key point: Mutually exclusive: P(A ∩ B) = 0, so P(A | B) = 0.

Some pairs of events can never happen together:

- A single coin toss cannot land both heads and tails.
- At a junction, a driver turns either left or right, not both.
- One die cannot show both 3 and 6 on the same roll.

Such events are **mutually exclusive** (G-1286): they have no outcome in common, so at most one of them can happen. As a formula, A and B are mutually exclusive if

$$P(A \cap B) = 0$$

Here $A \cap B$ (the **intersection**, G-967) is the event "both A and B happen", and $P(\cdot)$ is the probability of an event. For one die roll, with A = "shows 3" and B = "shows 6", no roll is in both, so $P(A \cap B) = 0$.

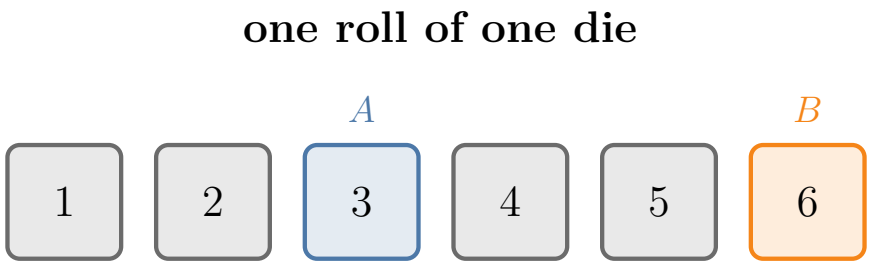


Figure 1: One roll of one die, with A = "shows 3" and B = "shows 6". The two events share no face

Figure 1 shows the die example: A and B sit on different faces, so no outcome is in both. Once we know B happened, the roll is face 6, and face 3 is ruled out.

3. Conditional probability for mutually exclusive events

> **Key point:** If B happened, A cannot have happened: P(A | B) = 0.

The **conditional probability** $P(A \mid B)$ is the probability of A once we know B happened (the definition). Putting $P(A \cap B) = 0$ into its formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{P(B)} = 0$$

With numbers: die 1 has shown 3. The probability that the same die shows 6 on that roll is 0. A driver has turned left; the probability that they turned right is 0.

Figure 2 checks this on a million simulated rolls. Over all rolls, a 3 comes up 0.1671 of the time, close to the exact value 1/6. Among the rolls that showed 6, not one showed 3: the share is exactly 0. The right panel previews the **addition rule** (G-173) of Section 5.

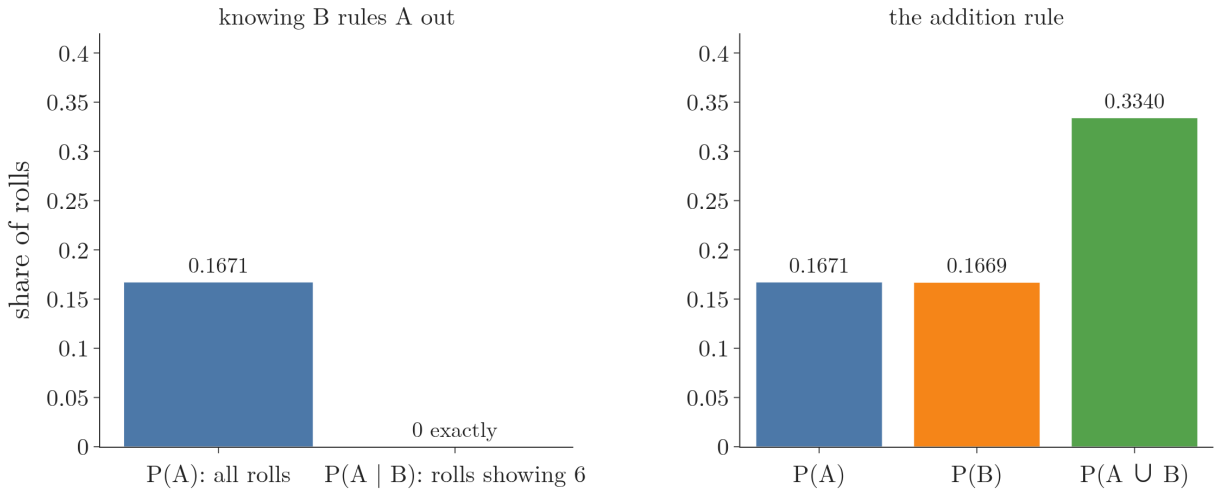


Figure 2: One million rolls of a fair die. Left: the share showing 3 is 0.1671 over all rolls, and exactly 0 among the rolls that showed 6. Right: the shares of 3, of 6, and of "3 or 6", which is their sum, about 1/3.

4. Mutually exclusive is not independent

> **Key point:** Knowing B happened tells us A did not, which is a lot of information. So **mutually exclusive events** (G-1286) (each with a positive probability) are always dependent.

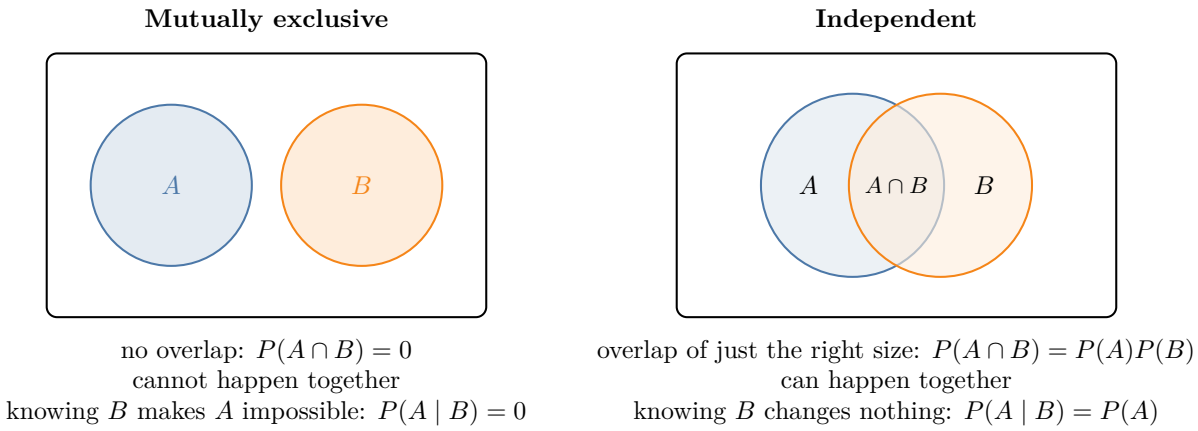


Figure 3: Mutually exclusive versus independent events

Figure 3 lays the two ideas side by side; the table below says the same in rows.

	Mutually exclusive	Independent
Can happen together?	no	yes
$P(A \cap B)$	0	$P(A) \times P(B)$
$P(A \mid B)$	0	$P(A)$
Does B tell us about A?	yes: A did not happen	no

For "die 1 shows 3" (A) and "die 1 shows 6" (B), each has probability 1/6:

- $P(A \cap B) = 0$, so they are mutually exclusive.
- The product of the two probabilities is not 0, so they are **not** independent:

$$P(A) \times P(B) = 1/36 \neq 0$$

In fact, if both events have a probability above 0, being mutually exclusive always rules out being independent. Learning that one happened tells us for certain that the other did not.

An everyday picture: a light switch is either on or off. The two states are mutually exclusive, and for that very reason, hearing "the switch is on" tells us everything about "off".

5. The addition rule: where mutual exclusivity pays off

> **Key point:** P(A or B) = P(A) + P(B) − P(A and B). For **mutually exclusive events** (G-1286) the last term is 0, so "or" becomes plain addition.

The probability that A **or** B happens (one, the other, or both) is the probability of their **union** (G-2045), written $P(A \cup B)$. Counting it needs care when the events share outcomes.

A bag holds 29 objects: 5 yellow cubes, 7 yellow spheres, 8 green cubes and 9 green spheres. One object is drawn at random. What is the probability that it is yellow or a cube?

1. **Count yellow:** 5 cubes and 7 spheres.

$$5 + 7 = 12$$

$$P(\text{yellow}) = 12/29$$

2. **Count cubes:** 5 yellow and 8 green.

$$5 + 8 = 13$$

$$P(\text{cube}) = 13/29$$

3. **Adding them counts some objects twice:** the 5 yellow cubes are in both counts.

$$12 + 13 = 25$$

4. **Subtract the overlap once:**

$$12 + 13 - 5 = 20$$

$$P(\text{yellow or cube}) = 20/29$$

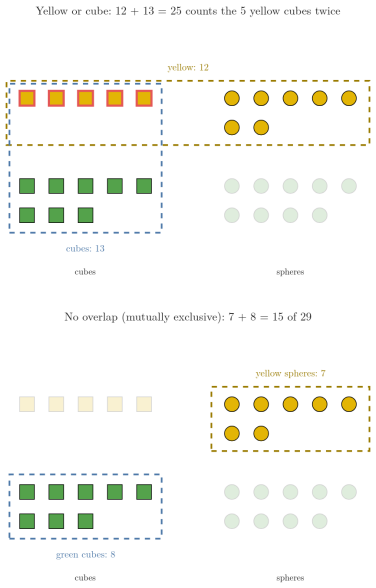


Figure 4: The bag of 29 objects. Yellow (12) and cube (13) share the 5 yellow cubes, so their counts add up to 25 with 5 counted twice; subtracting the 5 gives 20 of 29. Yellow spheres (7) and green cubes (8) share nothing, so their counts simply add: 15 of 29.

In Figure 4, watch the 5 yellow cubes (ringed in red): they lie inside both dashed boxes, so a plain sum counts them twice. Dividing every count in step 4 by 29 gives the general **addition rule**:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Now take two events with no overlap: "yellow sphere" (7 objects) and "green cube" (8 objects). No object is both, so the events are mutually exclusive and $P(A \cap B) = 0$. Nothing is counted twice (Figure 4, last scene), and the rule becomes a simple sum, the **addition rule for mutually exclusive events** (G-173):

$$P(A \cup B) = P(A) + P(B)$$

$$P(A \cup B) = \frac{7}{29} + \frac{8}{29}$$

$$P(A \cup B) = \frac{15}{29}$$

The die of Section 3 follows the same rule (Figure 2, right):

$$P(3 \text{ or } 6) = 1/6 + 1/6 = 1/3$$

Bayes' theorem (G-269) uses this sum to split a probability into separate cases.

6. Summary

- **Mutually exclusive** (G-1286): $P(A \cap B) = 0$; the events cannot happen together, and $P(A \mid B) = 0$, because once B happened, A is ruled out.
- Independent: $P(A \cap B) = P(A)P(B)$; they can happen together, and $P(A \mid B) = P(A)$, because knowing B tells us nothing about A.
- Mutually exclusive events with positive probabilities are never independent, because learning that one happened tells us for certain that the other did not. So the two ideas must not be mixed up.
- **Addition rule** (G-173): $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; for mutually exclusive events, $P(A \cup B) = P(A) + P(B)$, because nothing is counted twice. **Bayes' theorem** (G-269) uses this sum to split a probability into separate cases.

7. Sources

Built from

- CampusX, "Naive Bayes Classifier | Part 3 | Mutually Exclusive Events", YouTube, https://www.youtube.com/watch?v=nneTjTYikBE
- Khan Academy, "Addition rule for probability | Probability and Statistics", YouTube, https://www.youtube.com/watch?v=QE2uR6Z-NcU (the bag of 29 objects and the addition rule; Figure 4 recreates the count with our own drawing)

8. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Mutually exclusive events (G-1286)	Events that cannot happen at the same time; their intersection has probability 0.
Addition rule (general addition rule) (G-173)	The rule for the probability that A or B (or both) happens: add their probabilities and subtract the overlap so it is not counted twice, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; for mutually exclusive events the overlap is 0.
Union (A ∪ B) (G-2045)	The set of elements in A or B (or both), such as the points inside either of two convex pieces of an obstacle; for events, the event that A or B (or both) happens.
Independent events (G-934)	Events where one happening does not change the probability of the other; for them the probability that both happen is the product of their probabilities.

Term	Meaning
Intersection (A ∩ B) (G-967)	The set of elements that are in both A and B, such as the points inside both of two half-planes; for events, the event that both A and B happen.
Bayes' theorem (G-269)	The rule that reverses a conditional probability: from $P(B \mid A)$ it gives $P(A \mid B)$, so a belief about A can be updated after seeing B; $P(A \mid B) = P(B \mid A)P(A)/P(B)$.